# FORMAL SMOOTHNESS DESCENDS ALONG A RING MAP THAT IS A DIRECT SUMMAND

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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*Allowed standard prerequisites (editorial).*

- Formal smoothness in the adic topology, defined through the lifting of diagrams along square-zero ideals. the author cites Definition definition-formally-smooth at proof lines 9389-9392. Supplies the property to be descended and the form of the diagram that has to be lifted.
- Base change of a lifting diagram along the ring map R to R'. used at proof lines 9393-9400. Produces the diagram to which the hypothesis applies.
- A direct summand decomposition R' = R plus C and the resulting decomposition of the base changed target, with the complement part an ideal because the relevant ideal squares to zero. used at proof lines 9400-9416. Is the computation that produces the arrow R to S to A required by the definition.

*Source reference.* Original lemma 07EH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9447–9447. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 07EB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9441–9441. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $R$, $S$ be rings. Let $\mathfrak{n} \subset S$ be an ideal. Let $R \to R'$ be a ring map. Set $S' = S \otimes_R R'$ and $\mathfrak{n}' = \mathfrak{n}S$. If*

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(1) *the map $R \to R'$ embeds $R$ as a direct summand of $R'$ as an $R$-module, and*
(2) *$R' \to S'$ is formally smooth for the $\mathfrak{n}'$-adic topology,*

*then $R \to S$ is formally smooth in the $\mathfrak{n}$-adic topology.*

*Proof.* Let a solid diagram

$$\begin{array}{ccc} S & \longrightarrow & A/J \\ \uparrow & & \uparrow \\ R & \longrightarrow & A \end{array}$$

as in Definition 07EB be given. Set $A' = A \otimes_R R'$ and $J' = \mathrm{Im}(J \otimes_R R' \to A')$. The base change of the diagram above is the diagram

$$\begin{array}{ccc} S' & \longrightarrow & A'/J' \\ \uparrow & \overset{\psi'}{\swarrow} & \uparrow \\ R' & \longrightarrow & A' \end{array}$$

with continuous arrows. By condition (2) we obtain the dotted arrow $\psi' : S' \to A'$. Using condition (1) choose a direct summand decomposition $R' = R \oplus C$ as $R$-modules. (Warning: $C$ isn't an ideal in $R'$.) Then $A' = A \oplus A \otimes_R C$. Set

$$J'' = \mathrm{Im}(J \otimes_R C \to A \otimes_R C) \subset J' \subset A'.$$

Then $J' = J \oplus J''$ as $A$-modules. The image of the composition $\psi : S \to A'$ of $\psi'$ with $S \to S'$ is contained in $A + J' = A \oplus J''$. However, in the ring $A + J' = A \oplus J''$ the $A$-submodule $J''$ is an ideal! (Use that $J^2 = 0$.) Hence the composition $S \to A + J' \to (A + J')/J'' = A$ is the arrow we were looking for. $\square$